

Yash Kumar

📞 9827767252 ✉ yashkumar18ch@gmail.com 🔗 linkedin.com/in/yashkumar1801 🌐 yashkumar181 🌐 Portfolio

Education

Jaypee University of Engineering and Technology

2024 – Present

B.Tech in Computer Science

CGPA: 9.16

Experience

Shyara Tech Solutions

Jan 2026 – Apr 2026

Data Science Intern

- Worked on real-world data-driven applications, contributing to analytics, backend systems, and dashboard development.
- **Bitex Restaurant Management System:** Created structured testing datasets to simulate real-world restaurant operations. Conducted research on analytics features including order trends and revenue insights.
- **Financial Management Dashboard:** Built a WhatsApp chatbot to collect user financial data (expenses, investments, subscriptions). Independently designed and implemented the backend database system, and developed a dashboard to visualize financial insights and track spending patterns.

Projects

🌐 **YapZone: Real-Time Messaging Platform** | *Next.js, TypeScript, Convex, Tailwind CSS*

February 2026

- Built a real-time messaging application supporting direct messages and group chats using Next.js and TypeScript.
- Built the backend using Convex for synchronization and made a full-text message search via Convex search indexes.
- Developed advanced chat functionalities l read receipts, typing indicators, message reactions and delete for me/everyone.

🌐 **Hybrid Residual Learning for AQI Prediction** | *Python, TensorFlow, XGBoost, LSTM*

Feb 2026

- Built a **Hybrid Model** (XGB-LSTM) that combines Gradient Boosting for AQI Predictions.
- Achieved an **11.52% reduction in RMSE** (21.32) compared to standalone XGBoost models, specifically optimizing performance during volatile Severe pollution events.
- Built a pipeline processing chemical features (NO_x , Benzene, etc), using LSTM to capture dependencies in residual errors.

🌐 **Samriddhi: Credit Risk Assessment Engine** | *Python, XGBoost, SHAP, Scikit-learn*

Oct 2025

- Built a credit scoring classifier using **XGBoost**, using a **261,746-household NSSO dataset** to estimate income proxies for unbanked beneficiaries.
- Built a dual-model AI tool with XGBoost repayment model and income proxy estimator trained on NSSO data.
- Used **SHAP** to provide model interpretability to justify risk scores to non-technical stakeholders.
- Designed scoring algorithm with a **60:40 weighted** (Repayment Probability + Income Proxy) to put users into risk tier.
- Built APIs using FastAPI and integrated a Streamlit dashboard.

🌐 **Manova: Multi-Tenant Mental Health Platform** | *React, TypeScript, Firebase*

September 2025

- Built a full-stack, multi-tenant mental health platform with student , admin dashboards for institution-level deployment.
- Built a structured Firebase Firestore data model with collections for colleges, students, counselors, appointments, forum posts, resources, and mental health assessments.
- Made secure **Role-Based Access Control** using Firebase Authentication and Firestore security rules to strictly separate student and admin access.
- Integrated an **AI chatbot using Groq LLM API** with issue tagging and anonymized aggregation to surface campus-wide mental health trends.

Technical Skills

Languages: Python, JavaScript, C++, Java, SQL

Machine Learning: XGBoost, Scikit-learn, K-Means, SHAP

Web Development: Node.js, React, Next.js, FastAPI, Firebase

Tools: Git, GitHub, VS Code, Jupyter Notebook

Achievements

Winner, HACTRON 2025 Hackathon